

Drumil Patel

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EDUCATION

Indian Institute of Technology, Roorkee 06/2018 – 06/2022 | Roorkee, India
Bachelor of Technology, Electrical Engineering

PROFESSIONAL EXPERIENCE

Senior Member of Technical Staff 07/2024 – Present | Bangalore, India
Devrev

- Owned the architecture for DevRev's native observability warehouse (MELTS - Metrics, Events, Logs, Traces & Sessions), powering observability for enterprise customers including Bajaj, HDFC, ICICI, and Axis Bank, as well as internal use cases spanning agent evaluation, commerce, and incident management
- Ingestion**
 - Designed a horizontally scalable ingestion pipeline sustaining 35 TB/hour of compressed telemetry and 5-6 PB of monthly volume, replacing a legacy MongoDB/EBS architecture with a self-healing, elastic S3-based storage layer that decouples compute from storage and enables independent scaling of ingestion and persistence.
 - The compute/storage decoupling delivered a 5x increase in ingestion capacity while cutting infrastructure costs (\$100K → \$17K/month, ~\$1M annually) - trading fixed, always-on EBS provisioning for elastic, usage-based S3 storage
 - Extended the architecture with compliance-grade BYOB (Bring Your Own Bucket) and BYOK (Bring Your Own Key) primitives to support regulated enterprise customers
- Read**
 - Designed and built a distributed SQL query engine using Apache DataFusion and Ballista, integrating Tantivy for full-text search, Bloom filter-based indexes for trace search, and Parquet-based columnar storage on S3, enabling highly parallel query execution with low latency at PB scale using S3 Byte Headers, with materialized views for fast dashboard queries.
 - Built an extensible UDF framework supporting anomaly detection and log pattern mining (Drain3)
- Architected MELTS' downstream ecosystem extensibility layer, enabling a range of use cases built directly on the MELTS warehouse including commerce, incident management workflows, LLM/Harness observability and agent-improvement pipelines
- Tech Stack Used:** Golang, Rust, DataFusion Ballista, Fluentd, Tantivy, Parquet, S3

Platform Engineer 05/2022 – 06/2024 | Bangalore, India
Newton School, Incanus Technologies

- Developed control plane for learning environments similar to replit, Github Codespaces which can spawn interactable environments(pods) for students like Jupyter Notebooks, Nodejs + Express. **Tech Stack Used:** Go, K8s SDK, Crossplane.
- Designed an event-driven infrastructure with cutting edge for ML and AI production workloads for Kubernetes, seamlessly integrating DAPR and Kafka as universal event brokers. This architecture serves as the foundation for successfully deploying over 15+ models into production. **Tech Stack Used:** Kafka, DAPR, Flask, Gunicorn, Kubernetes

Software Engineering Intern 06/2021 – 07/2021 | Hyderabad, India
Amazon

- Build a centralized configuration store for maintaining uniformity of configurations across all FBA Export Services using Amazon's internal Tree based database, with custom java library for extraction of configuration based on service locations.
- Tech Stack Used:** Java, Spring boot, Amazon Web Services, G2S2(Amazon Internal Tree based data structure)

Prometheus, Google Summer of Code 05/2020 – 08/2020 | Remote
Cloud Native Computing Foundation

- Added EKS-cluster support to Prombench, Prometheus's Kubernetes-based benchmarking control plane, orchestrating ~1,000 pods across scrapers and readers to stress-test Prometheus performance at scale (GSoC blog 🔗)
- Tech Stack Used:** Golang, Kubernetes, Prometheus, AWS, GCP

Submittly, Google Summer of Code 05/2019 – 08/2019 | Remote
Rensselaer Polytechnic Institute, New York

- Implemented randomized input/output generation to prevent hardcoded solutions and ensure authentic assessment, while engineering a RabbitMQ-backed, worker-driven evaluation infrastructure capable of processing **10,000 students per hour** and **60,000 submissions during exam windows**. **Tech Stack Used:** Python, C++, RabbitMQ, PostgreSQL, Symphony

PUBLICATIONS

Random Input and Automated Output Generation in Submittly 🔗 2020
ACM Technical Symposium on Computing Science Education (SIGCSE)